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Changes in mechanical load and extensor muscle activity in the cervico-thoracic spine induced by sitting posture modification

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The influence of whole body sitting posture on cervico-thoracic posture, mechanical load and extensor muscle activity was examined in 23 asymptomatic adults. Cervical and upper thoracic extensor muscle activity measured in guided slouched and lumbo-pelvic neutral postures was normalised to that measured in a self-selected habitual posture. Head and neck posture and gravitational load moment measurements were obtained in each posture. Sagittal head translation, upper cervical extension and load moment were significantly greater in the slouched posture ($p < 0.001$). Contrasting patterns of cervical and thoracic extensor activity were observed in the slouched and neutral postures, with cervical extensor activity 40% higher in the slouched posture ($p < 0.0001$). Thoracic extensor activity was significantly higher in the lumbo-pelvic neutral posture than the habitual posture ($p = 0.002$). The significant changes in extensor muscle activity with postural modification appear to be induced by the associated change in mechanical load moment of the head.

Statement of Relevance: More neutral sitting postures reduce the demand on the cervical extensor muscles and modify the relative contribution of cervical and thoracic extensors to the control of head and neck posture. Postures that promote these patterns of muscular activity may reduce cervical spine loading and the development of posture-related neck pain.

Keywords: cervical spine; cervico-thoracic muscle activity; electromyography; sitting posture

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1. Introduction

Sustained sitting postures and the related load on the cervical spine are important contributors to the high prevalence of neck pain in office workers and individuals performing computer-based tasks (Côté *et al*. 2008). The loads transferred to the neck are likely to be influenced by the posture of the head and neck and related levels of muscle activity (Choi 2003). Flexed head and neck postures are associated with an increase in gravitational load moment and cervical extensor muscle activity, which may contribute to the higher prevalence of neck pain in individuals with this postural habit (Yip *et al*. 2008, Straker *et al*. 2009). Similarly, correction towards a more upright posture tends to decrease cervical extensor activity and increase activation of the deep flexor muscles (Falla *et al*. 2007, Caneiro *et al*. 2010).

Changes in whole body sitting posture have been shown to influence head and neck posture and levels of activity in the neck flexor and extensor muscles (Black *et al*. 1996, McLean 2005, Falla *et al*. 2007). Sitting postures that promote cervico-thoracic flexion are associated with greater cervical extensor muscle activity, while more upright sitting postures that reduce forward head translation and cervical flexion appear to reduce the level of cervical extensor activity (Harms-Ringdahl *et al*. 1986, Caneiro *et al*. 2010). Changes in sitting posture influence the alignment of both the cervical and thoracic regions (O'Sullivan *et al*. 2006, Edmondston *et al*. 2007). The associated changes in cervical extensor muscle activity have been previously reported (Thuresson *et al*. 2005), but the associated changes in thoracic extensor activity are less clear.

Cervical spine extensor muscle activity is likely to vary according to the load demands induced by the position of the head and neck. The loading demands increase as the head and neck are more flexed, increasing the gravitational load moment around the seventh cervical vertebra (Straker *et al*. 2009). The influence of cervical load moment on extensor muscle activity in functional postures has been examined in only a limited number of studies. In helicopter pilots, a 20° increase in cervical flexion was associated with a three-fold increase in cervical load moment, but a very small increase in neck extensor muscle activity (Thuresson *et al*. 2005). The limited change in muscle activity was despite the

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pilots wearing a helmet and night vision goggles, which would increase the load moment relative to that induced by the weight of the head alone. However, Harms-Ringdahl *et al*. (1986) reported a similar increase in cervical load moment when sitting postures were modified from neutral to end-range flexion. However, the associated median extensor muscle activity was 6% maximum voluntary contraction (MVC), although higher levels (10% MVC) were recorded in some subjects. The relationship between cervical load moment and extensor muscle activity has not been examined with regard to mid-range cervical postures and the influence of whole body postural adjustment.

Sitting postures are typically associated with levels of neck muscle activation of between 2 and 6% of MVC and changes in response to postural adjustment are similarly small (McLean 2005, Mork and Westgaard 2006). Normalisation of neck muscle electromyography (EMG) using a MVC is typically employed but MVC can be influenced by factors such as subject motivation and the posture in which the maximal contraction is performed (Mathiassen *et al*. 1995). Although postural modification has been associated with significant changes in neck extensor muscle activity, the magnitude of this change is commonly less than 5% of MVC (McLean 2005, Thuresson *et al*. 2005, Caneiro *et al*. 2010). Therefore, in studies examining postural neck muscle activity, EMG normalisation using a reference posture, has been recommended (Sommerich *et al*. 2000). This approach may provide a better appreciation of the relative changes in cervical and thoracic muscle activity in response to postural modification. Therefore, the objective of this study was to examine the influence of whole body sitting posture on cervico-thoracic posture and the gravitational load moment around an axis in the low cervical spine induced by the weight of the head. A further objective was to measure the posture-related changes in cervical and thoracic extensor muscle activity, where muscle activity in each posture was referenced to that measured in the habitual sitting posture.

2. Methods

2.1. Subjects

Altogether, 23 asymptomatic subjects (11 males and 12 females) with a mean age of 27.7 years (SD 3.5) were recruited for the study. Subjects had no history of neck pain in the last 3 months and no restriction of cervico-thoracic range of motion. Subjects were excluded if they had pain or medical conditions affecting cervical mobility, neurological symptoms affecting the upper limb, arm pain or pain associated with upper limb movement, previous neck trauma or were currently receiving treatment. All subjects signed a written consent form following a detailed explanation of the study procedures. Approval for the study was granted by the institutional Human Research Ethics Committee.

2.2. Experimental procedure

A within subjects, repeated measures design was employed to examine the influence of postural change on cervico-thoracic load moment and muscle activity as measured using surface EMG. Three postures were examined, which were modified from those described by Burnett *et al*. (2009):

- Habitual posture: Subjects were asked to sit in a relaxed position on a treatment couch without any specific instruction as to how they should sit (Figure 1a). To facilitate relaxation into a habitual posture, subjects were engaged in conversation to distract their attention from their posture. This position was maintained for a minimum of 3 min prior to posture analysis.
- Lumbo-pelvic neutral posture: Subjects were guided into a neutral lumbo-pelvic posture. This posture required anterior rotation of the pelvis, a neutral lumbar lordosis and positioning of the thorax over the pelvis but without excessive extension (Figure 1b).
- Slouched posture: Subjects were guided into a position of end-range lumbo-pelvic flexion, which induced thoracic flexion and cervical spine and head protraction (Figure 1c).

For all postures examined, subjects sat on a treatment couch with no back support. The height of the couch was adjusted to accommodate 90° of knee flexion with the feet positioned on the floor, shoulder width apart. Arms were relaxed with the hands resting palm down on the thighs. In all postures, subjects were asked to fix their gaze on an adjustable wall marker, which was positioned at eye level. The participants were guided into the prescribed postures by an experienced physiotherapist. Subjects maintained each posture for 10 s to allow for posture analysis photographs and EMG measurements to be completed. Subjects were given 1 min rest between postures and each posture was tested in a random order.

3. Measurements

3.1. Electromyography

Following skin preparation of shaving and alcohol rub, bipolar surface electrodes (3M, Ag/AgCl

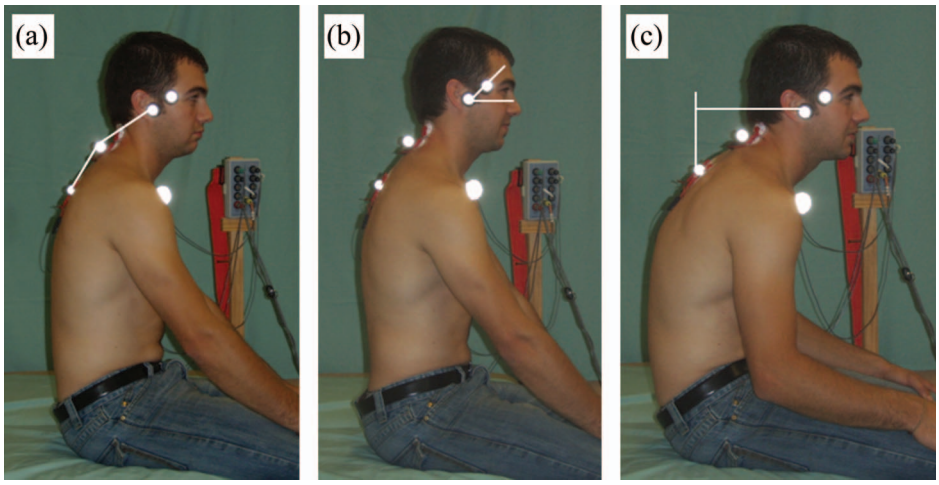


Figure 1. The habitual posture with cervico-thoracic angle (a), the lumpo-pelvic neutral posture with upper cervical angle (b) and the slouched posture with sagittal head translation (c).

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2248) were placed bilaterally according to anatomical location specific sites, as recommended by Sommerich *et al.* (2000). Electrodes were placed bilaterally over the cervical and thoracic erector spinae muscles. The cervical electrodes were positioned 2 cm lateral to the C4 spinous process and the thoracic electrodes positioned 5 cm lateral to the T4 spinous process. A reference electrode was placed over the acromion process. All inter-electrode impedance was less than 20 kΩ. In each posture, 10 s EMG signal was collected at a sampling rate of 1 kHz using an Octopus Cable Telemetric system (Bortec Electronics Inc., Calgary, Canada) with a 16 bit digital acquisition card (National Instruments, Austin, TX, USA) and a customised software program (LabVIEW V7.0; National Instruments).

All raw EMG signals were pre-amplified and visually inspected for artefacts (heart rate and movement). For each channel, individual amplification gains were set to ensure appropriate amplification without the potential for amplitude clipping. Post-processing included demeaning and secondary band-pass filtering (fourth order zero lag Butterworth filter 20–500 Hz). The EMG signal amplitude for each channel was determined using a total root mean square (RMS) for each second of the 10 s task duration. EMG signals were amplitude normalised (RMS) using a sub-maximal reference signal recorded during the habitual sitting posture, according to the recommendations for examining low levels of neck muscle activation (Sommerich *et al.* 2000). The normalised mean amplitude over the 10 s data collection period was used in the analysis.

3.2. *Posture measurements*

Reflective markers were adhered unilaterally over the C7 and T4 spinous processes, tragus of the ear, lateral margin of the orbit and sternal notch. An experienced physiotherapist identified the anatomical landmarks for marker placement and these were checked by a second investigator.

A digital camera (Sony DCR-SX403; Sony Corp., Tokyo, Japan) was mounted on a tripod and positioned 2.5 m to the left side of the subject. Photographs (2592 × 1944 pixels) were taken in each test position by the same investigator. The images were analysed on a personal computer using image analysis software (Image J; National Institutes of Health, Bethesda, MD, USA). All image analysis was performed by one investigator in order to minimise image analysis error. The postural measurements were the following:

- Cervico-thoracic angle: The angle formed by the intersection of lines connecting the tragus and the C7 spinous process, and the C7 spinous process and the T4 (Figure 1a).
- Upper cervical angle: The angle formed by a line extending from the tragus to the lateral margin of the orbit and the horizontal plane (Figure 1b).
- Sagittal plane head translation: The horizontal distance between the vertical line extending from the T4 spinous process and the tragus (Figure 1c).

3.3. *Cervico-thoracic load moment analysis*

The static cervico-thoracic load moment was calculated according to the method described by

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Harms-Ringdahl *et al*. (1986). To calculate the static load moment of the head and neck, the motion axis for C7/T1 was defined as the mid-point of a line extending between the C7 spinous process and the sternal notch. A perpendicular line was extended from the tragus (representing the line of gravitational force of the head) and the distance between this line and the motion axis at C7/T1 was calculated. The cervico-thoracic load moment was calculated using static mechanical analysis as:

$$M = F(h) \times d \quad (1)$$

where d is the perpendicular distance from the motion axis to the gravitational force of the head and F(h) is the weight of the head (defined as 7.9% of body weight) (Dempster 1955).

3.4. Posture measurement reliability

A reliability study was conducted on all posture measurements prior to commencing the study. Five images were acquired from one subject in each of the three postures. Each image was analysed by one investigator who was blinded to the measurement results. Coefficients of variation (CV) and standard error of measurement (SEM) were calculated for each posture. The CV was between 0.44% and 0.54% for the cervico-thoracic angle, 0.87% and 3.98% for upper cervical angle and 0.30% and 0.70% for anterior sagittal translation. The SEM for all postural measurements was $< 1^{\circ}$.

3.5. Statistical analysis

Descriptive statistic analysis was performed to summarise the posture and load moment measurements in each of the sitting positions. Since no significant difference in EMG amplitude between sides was identified, muscle amplitudes were averaged and the mean value used in all comparative analyses. The mean EMG amplitude expressed as change in amplitude (%) from the reference habitual posture amplitude was calculated for each prescribed posture. A one-factor repeated measures ANOVA was used to test for significant differences in the head and neck posture measures, and load moment, between the three sitting postures. Post hoc contrast analysis was then used to test for pair-wise differences. For the EMG data, a repeated measures ANOVA was used to determine if there was a significant effect of posture and muscle or a posture by muscle interaction. Post hoc pair-wise comparisons were then performed to test for significant differences in EMG amplitude between postures and muscles. For all analyses, the criterion for statistical significance was set at $p < 0.05$.

4. Results

4.1. Postural analysis

The posture measurements for each of the sitting postures are summarised in Table 1. There were no significant differences in the posture measures between the habitual and lumbo-pelvic neutral postures. The slouched posture was associated with a slight increase in the cervico-thoracic angle compared with the habitual and lumbo-pelvic neutral posture but the differences were not statistically significant ($p = 0.17$ and $p = 0.13$ respectively). Upper cervical extension was significantly greater in the slouched posture than in the habitual ($p < 0.0001$) or neutral ($p = 0.0004$) postures. Similarly, anterior translation of the head was significantly greater in the slouched posture than in the habitual ($p < 0.0001$) or neutral postures ($p < 0.0001$).

4.2. Cervico-thoracic load moment

The moment arm length and resultant gravitational load moment calculated in each posture are summarised in Table 1. The load moment in the slouched posture was significantly greater than in the habitual ($p < 0.0001$) and lumbo-pelvic neutral ($p < 0.0001$) postures. There was no significant difference in load moment between the habitual and lumbo-pelvic neutral postures. The load moment in the slouched posture increased by 57.2% compared with the habitual posture. There was a 2.2% decrease in the load moment in the lumbo-pelvic neutral posture compared with the habitual posture.

Table 1. Mean (SD) head and neck posture measures, moment arm length and load moment for each posture.

	Habitual	Slouched	Lumbo-pelvic neutral
Cervico-thoracic angle (°)	157.0 (6.1)	159.2 (8.7)	157.0 (6.9)
Upper cervical angle (°)	34.0 (12.9)[a]	23.8 (12.7)	31.6 (14.3)[b]
Sagittal head translation (mm)	193 (45)[a]	239 (52)	193 (52)[a]
Moment arm length (mm)	57 (22)[b]	91 (27)	57 (27)[b]
Cervical load moment (Nm)	3.1 (1.6)[b]	4.9 (2.1)	3.1 (1.9)[b]

[a]Significantly different to slouched ($p < 0.001$).
[b]Significantly different to slouched ($p < 0.0001$).

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4.3. Cervico-thoracic muscle activity

There was a significant interaction between posture and muscle group ($F22,1 = 7.55$, $p = 0.011$). The posture-related changes in cervical and thoracic muscle activity relative to the habitual posture are illustrated in Figure 2. In the slouched posture, there was a significant increase in cervical extensor muscle activity ($p < 0.0001$), but little change in the thoracic extensor activity. In contrast, there was a significant increase in thoracic extensor activity in the lumbo-pelvic neutral posture compared with the habitual posture ($p = 0.002$), but no change in the cervical extensor activity.

Differences in cervical and thoracic muscle activity in the slouched and lumbo-pelvic neutral postures are illustrated in Figure 3. In the slouched posture, cervical extensor activity was higher than the thoracic extensor activity but the difference was not significant ($p = 0.07$). Conversely, thoracic extensor activity was higher than cervical extensor activity in the lumbo-pelvic neutral posture but the difference was not significant ($p = 0.06$).

5. Discussion

Guidelines for computer and workstation use recommend lumbo-pelvic support to assist the maintenance of optimal head and neck postures (Cook and Burgess-Limerick 2003). This study has shown that modification of the lumbo-pelvic position in unsupported sitting has a significant influence on the posture of the head and neck. More upright lumbo-pelvic postures significantly reduce anterior translation of the head, and upper cervical extension, which are features of the forward-head posture, associated with an increased prevalence of neck pain (Aaras *et al*. 1998, Yip *et al*. 2008). This postural adjustment resulted in a significant decrease in the gravitational load moment around the low cervical spine, induced by the weight of the head and neck. While previous studies have reported significant increases in the load moment around the low cervical spine in mid-range and extreme neck flexion (Finsen 1999, Thuresson *et al*. 2005), the present results demonstrate a similar effect of neck protraction, a posture more commonly associated with computer use by office workers (Szeto *et al*. 2002, Yoo and Kim 2010).

Under low load conditions, posture of the head and neck is controlled predominantly by the deep anterior and posterior muscles of the neck (Mayoux-Benhamou *et al*. 1997). Changes in neck flexor and extensor muscle activity in response to whole body postural adjustments in sitting have previously been demonstrated (McLean 2005, Falla *et al*. 2007, Caneiro *et al*. 2010). In the present study, the level of activation of the cervical and thoracic extensor muscles appeared to be driven by the magnitude of the load moment of the head and neck. The higher load moment in the slouched posture was associated with significantly greater cervical extensor muscle activity than that associated with the lumbo-pelvic neutral posture. Importantly, the activation of the cervical extensor muscles in the more protracted head position was 40% greater than in the more upright habitual and lumbo-pelvic postures. These higher levels of extensor muscle activity may disturb the balance between

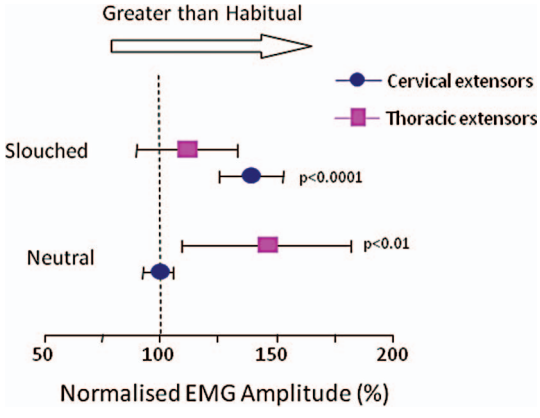


Figure 2. The mean electromyographic (EMG) amplitude of the cervical and thoracic extensor muscles normalised to the habitual posture. Cervical extensor activity was significantly greater in the slouched posture and thoracic extensor activity was significantly greater in the lumbo-pelvic posture.

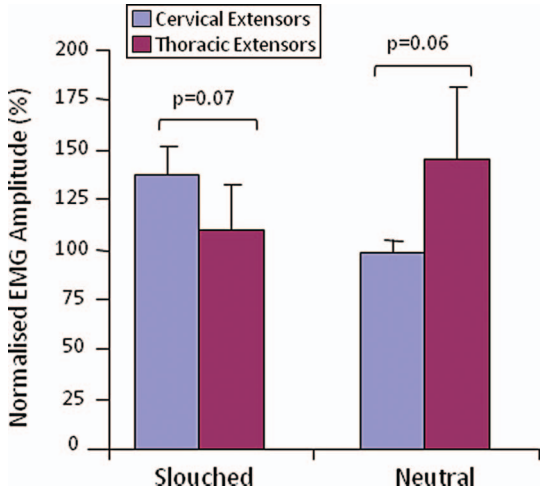


Figure 3. Comparison of cervical and thoracic extensor muscle activity in the lumbo-pelvic neutral and slouched postures normalised to electromyographic (EMG) activity in the same muscles in the habitual posture.

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extensor and flexor muscle activity, changing the optimal patterns and magnitude of load transfer in the cervical spine.

Previous studies examining muscle activity during sitting tasks have normalised the EMG measurements using maximal contraction in order to compare levels of activity in different muscles (Harms-Ringdahl *et al.* 1986, McLean 2005). The rationale for this is that %MVC gives a better indication of the relative demand on the muscle and its capacity (Straker *et al.* 2009). Given the low muscular demands of static sitting postures, the posture-related changes in cervical extensor activation relative to the MVC are typically very low, making the measurement vulnerable to error due to variability in the MVC (Jensen *et al.* 1996). The approach taken in the present study was to use a posture-specific normalisation technique so that EMG measurements in the prescribed postures were normalised to those in the subject's self-selected (habitual) posture (Bauer and Wittig 1998). As a result, cervical extensor muscle activity was about 40% greater in the slouched posture than in the habitual and lumbo-pelvic neutral postures. In previous studies, similar postural adjustments have been associated with changes in cervical extensor activity of between 1 and 5%MVC (McLean 2005, Caneiro *et al.* 2010). This suggests that EMG normalisation to a reference posture may provide a better indication of the relative changes in cervical muscle activity associated with changes in sitting posture.

While the changes in cervical extensor muscle activity associated with changes in head and neck posture are well documented, the changes in other related muscles are less clear. Contrasting patterns of thoracic and cervical extensor muscle activity were an unexpected finding of the present study. Upper thoracic extensor activity increased significantly in the lumbo-pelvic neutral posture compared with that in the habitual posture, despite the small changes in posture measurements and related load moment. This finding is consistent with that of Harms-Ringdahl *et al.* (1986), who reported a 2%MVC increase in upper thoracic extensor activity in a retracted head and neck posture, compared with that measured in a neutral posture. The cervical and thoracic extensor muscles may therefore have different roles in maintaining head and neck posture and resisting the flexion moments induced by the weight of the head. The upper thoracic extensors such as semi-spinalis cervicis and splenius cervicis may work to control flexion and protraction of the upper thoracic and low cervical spine through their cervical attachments. This would explain the significantly higher thoracic extensor activity in this muscle group in the more upright lumbo-pelvic posture and is consistent with the proposed postural role of these muscles (Conley *et al.* 1995). Despite the significant increase in thoracic muscle activity in the lumbo-pelvic neutral posture, there was no significant change in the posture measurements compared with those in the habitual posture. The likely explanation for this is that cervico-thoracic angle measurement used in this study was not sensitive to these subtle postural adjustments.

The contrasting patterns of cervical and thoracic muscle activity observed in this study provide an insight into optimal patterns of cervico-thoracic extensor muscle activity and how changes associated with forward head postures may lead to the development of postural neck pain. In the more upright lumbo-pelvic neutral posture, the increased activity in the upper thoracic extensors maintains a more retracted head position and a relatively low level of cervical extensor activity. The cervical lordosis would be further supported in this posture by increased activity in the deep anterior neck muscles (Fountain *et al.* 1966, Falla *et al.* 2007). In the forward head posture, decreased activation of the thoracic extensors is associated with a significant increase in cervical extensor activity. The increased activation in the cervical extensors, which have a smaller cross-sectional area compared to the extensors with a thoracic origin (Elliott *et al.* 2007), may result in decreased oxygenation and decreased force output in the low threshold motor units of these muscles (Kallenberg *et al.* 2007, Flodgren *et al.* 2009). Activation of higher threshold motor units may help off-set this fatigue response in the short term (Hodson-Tole and Wakeling 2009) and may explain the higher levels of muscle activity associated with low-load tasks, observed in some individuals with posture-related neck pain (Johnston *et al.* 2008). However, the higher levels of muscle activity would contribute to elevated levels of muscle strain and loading of cervical spine structures, increasing the potential for the development of pain.

Inferences drawn from the results of this study should acknowledge that the subjects were young and asymptomatic so that the findings may not be transferable to older individuals or those with neck pain. The study was conducted in unsupported sitting in order to evaluate muscular responses to postural change without the influence of external support. Although there was considerable variability in the habitual posture, the mean habitual posture used as the reference for the two prescribed postures was not significantly different from the lumbo-pelvic neutral posture. Subjects were encouraged to relax into their preferred posture prior to the photographic analysis. However, in some cases, there may have been a tendency to adopt a posture that was different to their true unsupported posture. The short duration of

posture analysis limits the extrapolation of these results to muscular responses to postures sustained over longer periods of time.

6. Conclusion

Work-related sitting postures that support the lumbo-pelvic region in a neutral position promote greater activation of the thoracic extensor muscles and significantly reduce the levels of cervical extensor muscle activity associated with a forward head posture. These posture-related changes in extensor muscle activity appear to be related to the position of the head relative to the trunk and changes in the gravitational load moment induced by the weight of the head. These observations will further inform the development of guidelines for optimal sitting support and postural alignment during occupational tasks.

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